SUBJECTIVE

2marks:

1. **Define Blastocoel?**

A fluid-filled cavity, the blastocoel, forms in the animal hemisphere. This cavity will be important for allowing cell movements to occur during gastrulation.

2. **Define periosteum?**

The entire region of calcified spicules becomes surrounded by compact mesenchymal cells that form the **Periosteum** (a membrane that surrounds the bone).

3. **Define Germ ring?**

After the blastoderm cells have covered about half the zebra fish yolk cell, a thickening occurs throughout the margin of the epibolizing blastoderm. This thickening is called the **germ ring**, and it is composed of a superficial layer, the **epiblast**, and an inner layer, the **hypoblast**.

4. **What is primitive knot or hensen node?**

At the anterior end of the primitive streak is a regional thickening of cells called the **primitive knot** or **Hensen's node.**

5. **Write difference between Bicoid and Nanos?**

The anterior most portion of the egg contains an mRNA that encodes a protein called Bicoid. The posterior most portion of the egg contains an mRNA that encodes a protein called Nanos.

6. **Difference between axoneme & dyenin?**

Answer: The major motor portion of the flagellum is called the **axoneme.**

The force for sperm propulsion is provided by **dynein**, a protein that is attached to the microtubules.

7. **Define Silencers?**

Enhancers can also be used to inhibit transcription. In some cases, the same transcription factors that activate the transcription of one gene can be used to repress the transcription of other genes. These "negative enhancers " are also called silencers.

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8. **What is embryonic shield?**

The cells of both the epiblast and hypoblast intercalate on the future dorsal side of the embryo to form a localized thickening, the embryonic shield.

9. **What is naïve?**

An uncommitted cell can be described as naive, meaning that it has received no instructions directing it along a particular developmental pathway.

10. **Egg jelly formation?**

Many types of eggs have glycoprotein meshwork called egg jelly outside the vitelline envelope which is used either to attract or to activate sperm. The egg, then, is a cell specialized for receiving sperm and initiating development.

11. **Differentiate between Blastomere and cleavage?**

Cleavage is a series of extremely rapid mitotic divisions wherein the enormous volume of zygote cytoplasm is divided into numerous smaller cells. These cells are called **blastomeres.**

12. **What is marginal belt?**

Between the area pellucida and the area opaca is a thin layer of cells called the **marginal zone** (or **marginal belt**).

13. **Define Neural Crest?**

A group of ectodermal cells that gives rise to the spinal ganglia and various structures of the autonomic nervous system.The neural crest cells originate at the dorsal most region of the neural tube.The neural crest can be divided into four main functional (but overlapping) domains.

14. **Define Regulative development?**

conditional specification gives rise to a pattern of embryogenesis called **regulative development**.

15. **Neural plate in tunicats?**

In **primary neurulation**, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.

16. **Role.of TGF nd FGF7 in human body development**

Paracrine factors such as **TGF- and FGF7** are important in normal skin development.

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17. **Define primary and secondary neurulation?**

In **primary neurulation**, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.

In **secondary neurulation**, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube. **Neurulation in fishes is exclusively secondary.**

18. **Define Prechordal plate?**

Cells migrating through Hensen's node become chordamesoderm (notochord) cells. These extend up to the presumptive midbrain, where they meet the prechordal plate. The **prechordal plate** induces the formation of the forebrain.

3MARKS

19. **Write three distinct lineage of Osteogenesis? Or What are lineages to generate skeleton?**

There are three distinct lineages that generate the skeleton.

- The somites generate the axial skeleton.
- The lateral plate mesoderm generates the limb skeleton.
- The cranial neural crest gives rise to the branchial arch and craniofacial bones and cartilage.

20. **What are syncytiotrophoblast and its function?**

The multinucleated type of cell forms the **syncytiotrophoblast**.

The extra-embryonic mesoderm joins the trophoblastic extensions and gives rise to the blood vessels that carry nutrients from the mother to the embryo. The narrow connecting stalk of extra-embryonic mesoderm that links the embryo to the trophoblast eventually forms the vessels of the **umbilical cord**.

21. **Write the function of Metencephalon?**

The Myelencephalon eventually becomes the **Medulla oblongata**, whose neurons generate the nerves that regulate respiratory, gastrointestinal, and cardiovascular movements. The **Metencephalon** gives rise to the cerebellum, the part of the brain responsible for coordinating movements, posture, and balance.

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22. Pattern of embryonic cleavage?

Patterns of embryonic cleavage

Indeed, different organisms undergo cleavage in distinctly different ways. The pattern of embryonic cleavage particular to a species is determined by two major parameters:

1. The amount and distribution of yolk protein within the cytoplasm.
2. The factors in the egg cytoplasm that influence the angle of the mitotic spindle and the timing of its formation.
3. The yolk-rich pole is referred to as the **vegetal pole**; the yolk concentration in the **animal pole** is relatively low. The zygote nucleus is frequently displaced toward the animal pole. In general, yolk inhibits cleavage.

23. Write the role of calcium in the fusion of egg and sperm?

Upon fertilization, the intracellular calcium ion concentration of the egg increases greatly. In this high-calcium environment, the cortical granule membranes fuse with the egg plasma membrane, releasing their contents. Once the fusion of the cortical granules begins near the point of sperm entry, a wave of cortical granule exocytosis propagates around the cortex to the opposite side of the egg. The release of calcium from intracellular storage can be monitored visually using calcium-activated luminescent dyes such as aequorin (isolated from luminescent jellyfish) or fluorescent dyes such as fura-2. These dyes emit light when they bind free calcium ions.

Several experiments have demonstrated that calcium ions are directly responsible for propagating the cortical granule reaction, and that these calcium ions are stored within the egg itself.

24. Differentiate between Autonomous and syncytial specification?

Autonomous specification:

Specification by differential acquisition of certain cytoplasmic molecules present in the egg.

- Characteristic of most invertebrates.
- Cell type specification precedes any large-scale embryonic cell migration.
- Autonomous specification gives rise to a pattern of development referred to as **mosaic development**, in which the embryo appears to be constructed like a tile mosaic of independent self-differentiating parts. Invariant cleavages produce the same lineages in each embryo of the species. Blastomere fates are generally invariant.

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Syncytial specification:

- Characteristic of most insect classes.
- Specification of body regions by interactions between cytoplasmic regions prior to cellularization of the blastoderm.
- Variable cleavage produces no rigid cell fates for particular nuclei.
- After cellularization, conditional specification is most often seen.

25. Difference between evolutionary embryology and tetralogy?

Evolutionary embryology:

The second strand, based on the first, is evolutionary embryology, the study of how changes in development may cause evolutionary changes and of how an organism's ancestry may constrain the types of changes that are possible.

Teratology:

The third anatomical approach to developmental biology is **teratology**, the study of birth defects. These anatomical abnormalities may be caused by mutant genes or by substances in the environment that interfere with development. The study of abnormalities is often used to discover how normal development occurs.

26. Difference b/w identical and fraternal twins?

Identical twins occur in roughly **0.25%** of human births. About **33%** of identical twins have two complete and separate chorions, indicating that separation occurred before the formation of the trophoblast tissue at **day 5.**

Fraternal twins are the result of two separate fertilization events, whereas identical twins are formed from a single embryo whose cells somehow dissociated from one another.

27. Differnce b/w gray matter and white matter?

Glial cells cover many of the axons in the marginal zone in myelin sheaths, giving them a whitish appearance. Hence, the mantle zone, containing the neuronal cell bodies, is often referred to as the **Gray matter**;

The axonal, marginal layer is often called the **White matter**. In the spinal cord and medulla, this basic three-zone pattern of ependymal, mantle, and marginal layers is retained throughout development. The gray matter (mantle) gradually becomes a butterfly-shaped structure surrounded by white matter; and both become encased in connective tissue.

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28. Define Medial hinge point?

In birds and mammals, the cells at the midline of the neural plate are called the **medial hinge point** (**MHP**) **cells**. They are derived from the portion of the neural plate just anterior to Hensen's node and from the anterior midline of Hensen's node.

29. Diff b/w late and early response of egg to sperm?

Early responses: Contact b/w sea urchin sperm & egg activates the 2 major blocks to polyspermy: the fast block, initiated by sodium influx into the cell, & the slow block, initiated by the intracellular release of Ca+ ions. The activation of all eggs appears to depend on an increase in the conc. of free Ca+ ions within the egg. Such an increase can occur in 2 ways: Ca+ ions can enter the egg from outside, or Ca+ ions can be released from the endoplasmic reticulum within the egg.
Late responses: Shortly after the Ca+ ion levels rise in a sea urchin egg, its intracellular pH also increases. The rise in intracellular pH begins with a 2nd influx of Na+ ions, which causes a 1:1 exchange b/w sodium ions from the seawater & hydrogen ions from the egg. This loss of hydrogen ions causes the pH to rise. It is thought that the pH increase and the calcium ion elevation act together to stimulate new protein synthesis and DNA synthesis.

30. Write the general concept of fertilization?

Fertilization is the process whereby two sex cells (gametes) fuse together to create a new individual with genetic potentials derived from both parents.

Fertilization accomplishes two separate processes:

- Sex (the combining of genes derived from the two parents) and
- Reproduction (the creation of new organisms)
- Thus, the first function of fertilization is to transmit genes from parent to offspring, and the second is to initiate in the egg cytoplasm those reactions that permit development to proceed.

31. What is Marble bone disease?

Osteopetrosis or Marble bone disease

Osteopetrosis is a bone disease that makes bones abnormally dense and prone to breakage (fracture).

This disorder is characterized by increased bone density and abnormal bone growth.

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32. Estrogen and progesterone function?

Estrogen is a hormone that can instruct the liver to make and secrete the yolk proteins, which are then transported through the blood into the enlarging eggs in the ovary. The yolk is transported into the bottom portion of the egg.

progesterone, signals the egg to resume its meiotic division. This is necessary because the egg had been "frozen" in the metaphase of its first meiosis. When it has completed this first meiotic division, the egg is released from the ovary and can be fertilized.

33. What are resact and its function?

Answer: **Resact** is specific for ***A. punctulata*** and does not attract sperm of other species. *A. punctulata* sperm have receptors in their plasma membranes that bind resact and can swim up a concentration gradient of this compound until they reach the egg. Resact also acts as a **sperm-activating peptide**.

5MARKS:

34. Write the functional domains of neural crest?

- A group of ectodermal cells that gives rise to the spinal ganglia and various structures of the autonomic nervous system.
- The neural crest cells originate at the dorsal most region of the neural tube.
- The neural crest can be divided into four main functional (but overlapping) domains.
- The cranial (cephalic) neural crest, whose cells migrate dorso-laterally to produce the craniofacial mesenchyme that differentiates into the cartilage, bone, cranial neurons, glia, and connective tissues of the face.

35. life cycle of chlamydomonas?

the life cycle of *Chlamydomonas.* Which is usually haploid, having just one copy of each chromosome. The individuals of each species, however, are divided into two mating types: plus and minus. When a plus and a minus meet, they join their cytoplasm, and their nuclei fuse to form a diploid zygote. The flagella of two individuals twist around each other, enabling specific regions of the cell membranes to come together. These specialized regions contain mating type-specific components that enable the cytoplasms to fuse.

36. Describe primary neurulation?

Primary neurulation

- During primary neurulation, the original ectoderm is divided into three sets of cells.
- The internally positioned neural tube, which will form the brain and spinal cord.

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- The externally positioned epidermis of the skin.
- The neural crest cells. The neural crest cells form in the region that connects the neural tube and epidermis, but then migrate elsewhere; they will generate the peripheral neurons and glia, the pigment cells of the skin, and several other cell types.

37. Transcription domain?

Transcription factors have three major domains.

1. **DNA-binding domain**
2. **Protein-protein interaction domain**
3. ***Trans*-activating domain**

- **DNA-binding domain:**It recognizes a particular DNA sequence.
- **Protein-protein interaction domain:** It allows the transcription factor's activity to be modulated by TAFs or other transcription factors.
- ***Trans*-activating domain:** It activates or suppresses the transcription of the gene whose promoter or enhancer it has bound. Usually, this *trans*-activating domain enables the transcription factor to interact with proteins involved in binding RNA polymerase.Examples,***MITF:(***Microphthalmia-Associated Transcription Factor. The microphthalmia (MITF) protein is necessary for the production of pigment cells and their pigments.
- ***Pax6:*** Paired box gene 6. The Pax6 transcription factor, which is needed for mammalian eye, nervous system, and pancreas development, contains two potential DNA-binding domains.
- **Pdx1:** Pancreatic And Duodenal Homeobox 1 gene. It is specific for the pancreatic region of the endoderm.

38. 2 ways for the formation of neural tube?

Formation of the Neural Tube

There are two major ways of forming a neural tube

1) Primary neurulation 2) **Secondary neurulation**

- In **primary neurulation**, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.

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- In **secondary neurulation**, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube. **Neurulation in fishes is exclusively secondary.**
- In **amphibians**, such as ***Xenopus,*** most of the tadpole neural tube is made by **primary neurulation**, but the tail neural tube is derived from **secondary neurulation**.

39. Enlist 4 factors that cause movement in gene regulation?

The regulation of gene expression can be accomplished at several levels:

- **Differential gene transcription:** It regulates that which of the nuclear genes are transcribed into RNA.
- **Selective nuclear RNA processing:** It regulating which of the transcribed RNAs (or which parts of such a nuclear RNA) enter into the cytoplasm to become messenger RNAs.
- **Selective messenger RNA translation:** It regulates that which of the mRNAs in the cytoplasm become translated into proteins.
- **Differential protein modification:** It regulates that which proteins are allowed to remain or function in the cell.

40. Write modes of commitment?

Three basic modes of commitment are

1: Autonomous specification:

Specification by differential acquisition of certain cytoplasmic molecules present in the egg.

- Characteristic of most invertebrates.
- Cell type specification precedes any large-scale embryonic cell migration.
- Autonomous specification gives rise to a pattern of development referred to as **mosaic development**, in which the embryo appears to be constructed like a tile mosaic of independent self-differentiating parts. Invariant cleavages produce the same lineages in each embryo of the species. Blastomere fates are generally invariant.

Produces "mosaic" ("determinative") development: cells cannot change fate if a blastomere is lost.

2: Conditional specification:

- Characteristic of all vertebrates and few invertebrates.

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- Specification by interactions between cells. Relative positions are important.
- Variable cleavages produce no invariant fate assignments to cells.
- Massive cell rearrangements and migrations precede or accompany specification.

Capacity for "regulative" development: allows cells to acquire different functions.

3: Syncytial specification:

- Characteristic of most insect classes.
- Specification of body regions by interactions between cytoplasmic regions prior to cellularization of the blastoderm.
- Variable cleavage produces no rigid cell fates for particular nuclei.
- After cellularization, conditional specification is most often seen.

41. Define : Area opaca , Area pellicuda , sub-germinal cavity, marginal zone?

- Between the blastoderm and the yolk is a space called the **sub-germinal cavity**.
- The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the **area opaca**.
- The deep cells in the center of the blastoderm are shed and die, leaving behind a onecell- thick **area pellucida**.
- Between the area pellucida and the area opaca is a thin layer of cells called the **marginal zone** (or **marginal belt**).

42. Hammering experiment for role of nucleus in morphology?

Nuclear control of cell morphogenesis and the interaction of nucleus and cytoplasm are well studied in *Acetabularia.* This enormous single cell (2-4 cm long) consists of three parts:

- a cap,
- a stalk,
- a rhizoid.

The single nucleus of the cell resides within the rhizoid. The size of *Acetabularia* and the location of its nucleus allow investigators to remove the nucleus from one cell and replace it with a nucleus from another cell.

J.Hammerling exchanged nuclei between two morphologically distinct species, *A. mediterranea* and *A. crenulata* **J. Hammerling** found that when he transferred the nucleus from one species into the stalk of another species, the newly

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formed cap eventually assumed the form associated with the *donor* nucleus. Thus, the nucleus was seen to control *Acetabularia* development.

The formation of a cap is a complex morphogenic event involving the synthesis of numerous proteins, which must be accumulated in a certain portion of the cell and then assembled into complex, species-specific structures.

10 Marks:

43. Endochondral?

- Endochondral ossification involves the formation of cartilage tissue from aggregated mesenchymal cells, and the subsequent replacement of cartilage tissue by bone.
- The process of Endochondral ossification can be divided into five stages.
- First, the mesenchymal cells are committed to become cartilage cells. This commitment is caused by paracrine factors that induce the nearby mesodermal cells to express two transcription factors, Pax1 and **Scleraxis**.
- Pax1 and Scleraxis are the transcription factors are thought to activate cartilage-specific genes.
- Thus, Scleraxis is expressed in the mesenchyme from the sclerotome, in the facial mesenchyme that forms cartilaginous precursors to bone, and in the limb mesenchyme
- During the second phase of endochondral ossification, the committed mesenchyme cells condense into compact nodules and differentiate into **chondrocytes**, the cartilage cells. N-cadherin appears to be important in the initiation of these condensations.
- During the second phase of endochondral ossification, the committed mesenchyme cells condense into compact nodules and differentiate into **chondrocytes**, the cartilage cells
- In humans, the *SOX9* gene, which encodes a DNA-binding protein, is expressed in the precartilaginous condensations. Mutations of the *SOX9* gene cause camptomelic dysplasia, a rare disorder of skeletal development that results in deformities of most of the bones of the body.
- During the third phase of endochondral ossification, the chondrocytes proliferate rapidly to form the model for the bone. As they divide, the chondrocytes secrete a cartilage-specific extracellular matrix In the fourth phase, the chondrocytes stop dividing and increase their volume dramatically, becoming hypertrophic chondrocytes. These large chondrocytes alter the matrix they produce (by adding collagen X and more fibronectin) to enable it to become mineralized by calcium carbonate.
- The fifth phase involves the invasion of the cartilage model by blood vessels.
- **The hypertrophic chondrocytes die by apoptosis.** This space will become **bone marrow**. As the cartilage cells die, a group of cells that have surrounded the cartilage model differentiate into osteoblasts.
- The osteoblasts begin forming bone matrix on the partially degraded cartilage. Eventually, all the cartilage is replaced by bone. Thus, the cartilage tissue serves as a model for the bone that follows

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44. Explain Osteogenesis in detail?

Osteogenesis:

- Some of the most obvious structures derived from the paraxial mesoderm are bones.

There are three distinct lineages that generate the skeleton.

- The somites generate the axial skeleton.
- The lateral plate mesoderm generates the limb skeleton.
- The cranial neural crest gives rise to the branchial arch and craniofacial bones and cartilage.
- There are two major modes of bone formation, or **osteogenesis**, and both involve the transformation of a preexisting mesenchymal tissue into bone tissue.
- The direct conversion of mesenchymal tissue into bone is called **intramembranous ossification**. This process occurs primarily in the bones of the skull.
- In other cases, the mesenchymal cells differentiate into cartilage, and this cartilage is later replaced by bone. The process by which a cartilage intermediate is formed and replaced by bone cells is called endochondral ossification.

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